Experiment 11

K-Nearest Neighbors (KNN).

Program:

```
import numpy as np
import pandas as pd
df=pd.read_csv('Iris.csv')
df.info()
df.variety.value_counts()
df.head()

features=df.iloc[:,:-1].values
label=df.iloc[:,4].values

from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier

xtrain,xtest,ytrain,ytest=train_test_split(
features,label,test_size=.2,random_state=1)

model_KNN=KNeighborsClassifier(n_neighbors=5)
model_KNN.fit(xtrain,ytrain)

print(model_KNN.score(xtrain,ytrain))
print(model_KNN.score(xtest,ytest))

from sklearn.metrics import confusion_matrix
confusion_matrix(label,model_KNN.predict(features))

from sklearn.metrics import classification_report
print(classification_report(label,model_KNN.predict(features)))
```

Output

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
0   sepal.length   150 non-null    float64
1   sepal.width    150 non-null    float64
2   petal.length   150 non-null    float64
3   petal.width    150 non-null    float64
4   variety        150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB

variety
Setosa        50
Versicolor    50
Virginica     50

   sepal.length  sepal.width  petal.length  petal.width variety
0           5.1          3.5           1.4          0.2  Setosa
1           4.9          3.0           1.4          0.2  Setosa
2           4.7          3.2           1.3          0.2  Setosa
3           4.6          3.1           1.5          0.2  Setosa
4           5.0          3.6           1.4          0.2  Setosa
```

Scores:

```
0.9583333333333334
1.0
```

Confusion Matrix:

```
[[50  0  0]
 [ 0 47  3]
 [ 0  2 48]]
```

Classification Report:

```
              precision    recall  f1-score   support

      Setosa       1.00      1.00      1.00        50
  Versicolor       0.96      0.94      0.95        50
   Virginica       0.94      0.96      0.95        50

    accuracy                           0.97       150
   macro avg       0.97      0.97      0.97       150
weighted avg       0.97      0.97      0.97       150
```